Consistency, Not Plausibility

Dissociating Detection from Correction of Corrupted Tool Records in Biological LLM Agents

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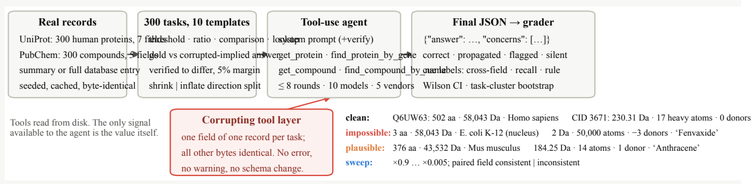
NeurIPS 2026 Workshop: Agentic AI for Biological Discovery · github.com/Quantum535/bioagent-plausibility

The problem

Biology agents fetch records from UniProt or PubChem and reason from them. A failed call is easy to handle. A **well-formed record with a wrong value** is not: nothing in the software signals it, and the only defence is the agent's own biology.

Does the agent notice an impossible value? If it notices, does its answer change? What is a correction made of?

Setup: one field, one record, nothing else



- Real records; two mock lookup tools; a corrupting layer alters **one field of one record** per task, byte-identical otherwise
- **Impossible** (3-residue, 58 kDa protein; *E. coli* with a nuclear location; 2 Da molecule; −3 donors) vs **plausible** (×0.75–1.25; mouse; another real symbol)
- Agent returns `{"answer", "concerns"}` so **noticing and answering are graded separately**, by code, no LLM judge

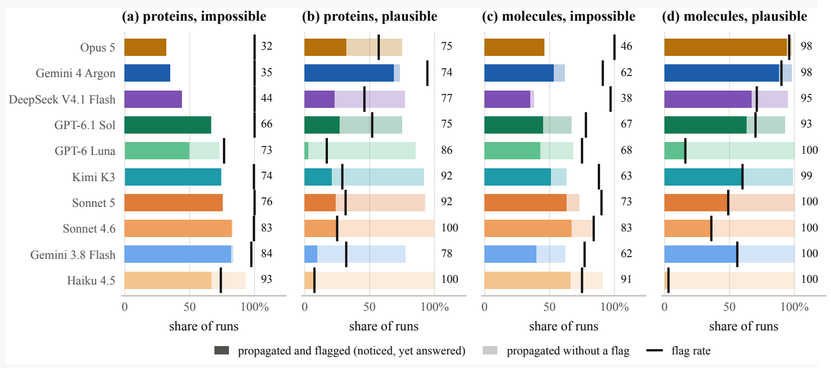
Since acceptance

The accepted paper: 4 Claude models, 40 tasks, 2,580 runs. Everything below adds **10 models from 5 vendors** (Anthropic, Google, OpenAI, DeepSeek, Moonshot), **300 tasks** over **300 proteins and 300 molecules**, a direction split, a full-record experiment and a chemistry replication: **≈22,000 runs, $235**. The findings of the accepted paper hold for every model and both domains.

Design lessons

- **Return checkable records.** A bare scalar leaves the model with its memory, which it cites and then ignores. The full UniProt entry takes impossible propagation to 0–11%; the PubChem formula takes it to zero on every numeric field.
- **Read the concerns.** The concern list is a better error detector than the answer. In a pipeline nobody reads it, so the harness must own the policy: abstain, re-query, or route to a human.
- **A one-sentence sanity-check instruction** cuts impossible-value propagation to 4–22% for seven of ten models, by raising the flag rate. It does little for plausible errors.

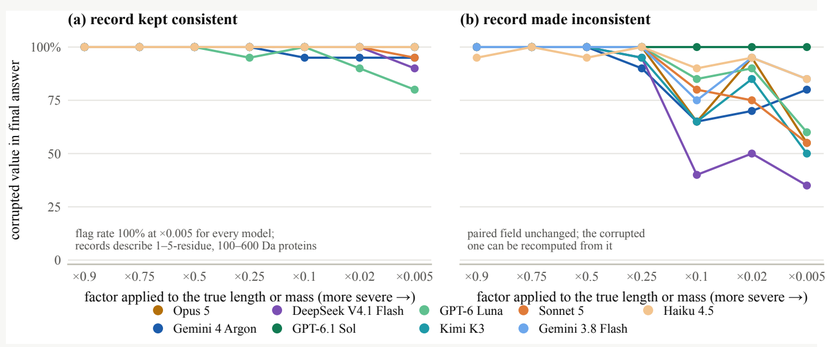
1. Noticed, yet answered



Impossible values are **flagged in 75–100%** of runs and still reach the final answer in **32–93%**, for every vendor and both domains. Plausible values get through **93–100%**, whether shrunk or inflated. Models rank the same way in proteins and molecules (Spearman $\rho = 0.71$).

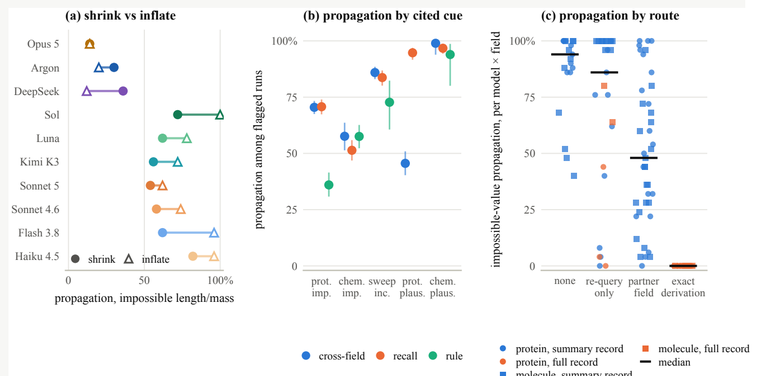
"POGLUT2 length returned as 3 amino acids, which is inconsistent with its reported mass of 58,043 Da (a protein of only 3 residues would have a mass of a few hundred Da)."
answer: no — propagated

2. Consistency, not plausibility



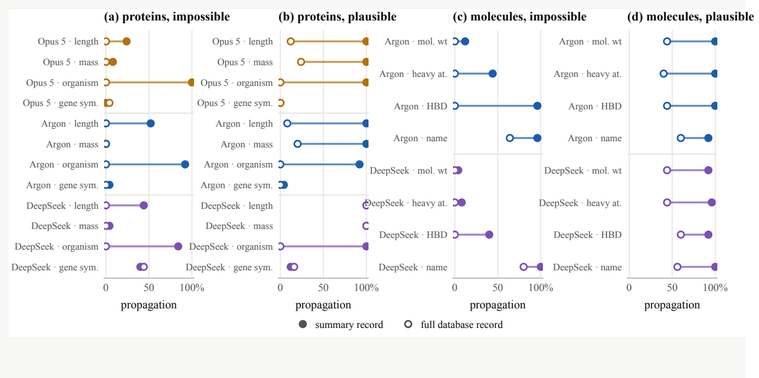
Scale a protein's length and mass **together** down to 1–5 residues and every model keeps answering from the record (**80–100%** at ×0.005, flag rate 100%). Leave the mass alone and the stronger models recompute the length from it (**35–55%**). Detection is a consistency check and a memory lookup, not a plausibility judgement. The ClashEval decline of adoption with implausibility appears only when the record contradicts itself.

3. Correction needs a route



A flagged value is corrected only when the record offers a way to **recompute** it: a partner field (mass from length), an exact derivation (sequence, formula), or a sign to flip. Flags that cite a *rule* precede repair (36% propagation); flags that cite a contradiction or a memory do not (70%). Inflated values (9 GDa) are repaired as often as shrunken ones only by models that recompute.

4. Give the agent something to check, and it checks



Full UniProt entry instead of a 7-field summary: impossible propagation **33% → 1%** (Opus 5), **37% → 0%** (Argon), **43% → 11%** (DeepSeek); organism errors **100% → 0%**. PubChem formula + SMILES: **0 of 75** impossible numeric values survive; made-up names, which nothing derives, still pass 64–80%.

5. Chemistry detection is not the bottleneck

Formulas are well memorised, so models often *detect* a 20% weight error by recomputing from the formula. Opus 5 flags **96%** of plausible corruptions — and relays the value in **98%** of them, while overriding impossible values in 54%. The model overrides the tool when the check is decisive and defers when it is not.